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SUBJECT CODE:24CSH-298

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Lab Experiment 5

AIM

To understand and apply conditional logic in SQL by using the modulus operator (MOD / %) to analyze numerical data and classify employee salaries as odd or even, thereby improving data analysis and decision-making skills in SQL.

Objective of the Session

- To write and execute SQL queries that use the MOD (%) operator to check employee salaries and display odd and even salary values separately from an employee table.

Software Requirements

Database Management System:

Oracle Database Express Edition (Oracle XE)

PostgreSQL Database

Database Administration Tool / Client Tool:

Oracle SQL Developer (for Oracle XE)

pgAdmin (for PostgreSQL)

Experiment Question :

Write SQL queries to classify employee salaries as **odd or even** using the **MOD (%) operator**.The structure of the table is as follows:

Practical/Experiment steps:

1. Open **Oracle SQL Developer / pgAdmin**.
2. Create an **Employee table** with employee details.
3. Insert sample employee salary records.
4. Execute SQL queries using the **MOD (%) operator**.
5. Display employees having **even salaries**.
6. Display employees having **odd salaries**.
7. Verify the output.

CODE:

```
-- Create Table
CREATE TABLE Employee (
emp_id INT,
emp_name VARCHAR(50),
salary INT
);

-- Insert Sample Data
INSERT INTO Employee VALUES (1, 'Aman', 25000);
INSERT INTO Employee VALUES (2, 'Riya', 30001);
INSERT INTO Employee VALUES (3, 'Rahul', 40000);
INSERT INTO Employee VALUES (4, 'Neha', 35005);

-- Display Employees with Even Salaries
SELECT *
FROM Employee
WHERE MOD(salary,2) = 0;

-- Display Employees with Odd Salaries
SELECT *
```

FROM Employee
WHERE MOD(salary,2) = 1;

OUTPUT:

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Query Query History

```
CREATE TABLE Employee (
    emp_id INT,
    emp_name VARCHAR(50),
    salary INT
);

INSERT INTO Employee VALUES (1, 'Aditya", 25000);
INSERT INTO Employee VALUES (2, 'Simran', 30001);
INSERT INTO Employee VALUES (3, ''Naveen', 40000);
INSERT INTO Employee VALUES (4, 'Priya', 35005);
```

Data Output Messages Notifications

```
INSERT 0 1

Query returned successfully in 36 msec.
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	emp_id integer	emp_name character varying (50)	salary integer
1	2	Simran	30001
2	4	Priya	35005

Successfully run. Total query runtime: 112 msec. 2

Total rows: 2 Query campete 00:00:0.112 CRLF Ln 12, Col

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	emp_id integer	emp_name character varying (50)	salary integer
1	1	Aditya	25000
2	2	Simran	30001
3	3	Naveen	40000
4	4	Priya	35005

Successfully run. Total query runtime: 55 msec. 4 rows

Total rows: 4 Query complete 00:00:0.055 CRLF Ln 12, Col

```sql
CREATE TABLE Employee (
    emp_id INT,
    emp_name VARCHAR(50),
    salary INT
);

INSERT INTO Employee VALUES (1, 'Aditya', 25000);
INSERT INTO Employee VALUES (2, 'Simran', 30001);
INSERT INTO Employee VALUES (3, 'Naveen', 40000);
INSERT INTO Employee VALUES (4, 'Priya', 35005);

SELECT * from Employee

SELECT *
FROM Employee
WHERE MOD(salary,2) = 0;

SELECT *
FROM Employee
WHERE MOD(salary,2) = 1;
```

Learning Outcome

1. Understand the use of the **MOD (%) operator** in SQL.
2. Apply **conditional logic** to analyze numerical data in a table.
3. Write SQL queries to **filter records based on conditions**.
4. Differentiate and display **odd and even salary values** from a database.